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3rd Sem / Computer, ECE, Automation & Robotics , Computer (For Speech and Hearing Impaired), ECE (For Speech and Hearing Impaired) , IT, Eltx, EI, , Med. Eltx., Power Eltx, Elect. & Eltx. Engg

Subject : Digital Electronics / Digital Eltx.- I / Digital Electronics & Computer Architecture

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Decimal number 15 in binary is:

a) 1110 b) 1100

c) 1111 d) 1001

Q.2 In digital systems, Gray code is mainly used to :

a) Detect errors

b) Convert decimal to binary

c) Simplify addition

d) Minimize errors during trasitions

Q.3 Truth table of OR gate gives output 1 when:

a) All inputs are 0 b) At least one input is 1

c) All inputs are 1 d) None of the above

Q.4 D flip-flop is also called:

a) Data or Delay flip-flop

b) Direct flip-flop

c) Differential flip-flop

d) Double flip-flop

Q.5 Universal shift register can:

a) Only shift left

b) Only shift right

c) Shift in both direction

d) Only store data

Q.6 Which memory can be erased by UV light?

a) PROM b) EPROM

c) RAM d) EEPROM

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 PROM_______.

Q.8 What is the symbol of XOR gate?

Q.9 Name any two logic gate.

Q.10 Name two types of counters.

Q.11 What is the function of an encoder?

Q.12 What is a flip-flop?

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 What is parity? Explain single and double parity.

Q.14 Write differences between TTL and CMOS logic families.

Q.15 Explain the working of half adder.

Q.16 What is demultiplexer? Design 16:1 Demultiplexer.

Q.17 Explain Decimal to BCD encoder with block diagram.

Q.18 Differentiate between Latches and Flip-Flop.

Q.19 Explain the performance characteristics of digital to analog convertor.

Q.20 Write short note on

a) RAM

b) ROM

Q.21 Compare synchronous and asynchronous counter.

Q.22 Write a short note on Excess-3 code and gray code.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 What do you mean by multiplexer? Design 32:1 MUX by using 16:1 MUX and 2:1 Mux

Q.24 Write short note on any two

a) latch

b) Encoder

c) Universal shift register

Q.25 Explain the working of master slave J-K flip-flop. How it will overcome the race around condition of flip-flop.